# 12-Week Practice Plan — Class 4 Mathematics Olympiad

Built around the SOF IMO pattern (35 Q / 40 marks / 60 min — see `02-exam-pattern.md`). Three phases: build core skills topic by topic, add pure logical reasoning, then stress-test with full timed papers.

Suggested pace: **3 short sessions a week, ~30 minutes each**, not one long cram — olympiad questions reward familiarity and speed more than raw study time.

## Phase 1 — Core Skills (Weeks 1–8)

One topic per week. Each week = 1 assignment of ~15 MCQs covering that topic's direct-computation + word-problem flavours (i.e. Mathematical Reasoning + Everyday Mathematics style questions together), plus a short self-check against the answer key.

| Week | Topic | Folder |
|---|---|---|
| 1 | Number Sense & Numerals (place value, expanded form, ordering, rounding, Roman numerals) | `assignments/01-number-sense/` ✅ |
| 2 | Four Operations (+ − × ÷, multi-digit, word problems) | `assignments/02-four-operations/` ✅ |
| 3 | Factors, Multiples & Number Patterns | `assignments/03-factors-multiples-patterns/` ✅ |
| 4 | Fractions & Decimals (comparing, equivalent, simple operations) | `assignments/04-fractions-decimals/` ✅ |
| 5 | Measurement: Length, Weight, Capacity + Money | `assignments/05-measurement-money/` ✅ |
| 6 | Time & Calendar | `assignments/06-time-calendar/` ✅ |
| 7 | Geometry (shapes, lines, symmetry) + Perimeter & Area | `assignments/07-geometry-perimeter-area/` ✅ |
| 8 | Data Handling (tally marks, pictographs, bar graphs) | `assignments/08-data-handling/` ✅ |

**Weekly routine:** attempt the assignment untimed first; mark it against the answer key; redo only the missed questions 2 days later to confirm the concept actually stuck.

## Phase 2 — Logical Reasoning (Weeks 9–10)

Pure Section-1 skills, since these don't show up anywhere else in the school curriculum and need dedicated practice.

| Week | Topic | Folder |
|---|---|---|
| 9 | Patterns, Analogy & Classification, Alphabet Test, Coding–Decoding | `assignments/09-reasoning-patterns-coding/` ✅ |
| 10 | Ranking Test, Direction Sense, Days & Dates, Mirror Images, Embedded Figures, Geometrical Solids | `assignments/10-reasoning-direction-figures/` ✅ (24 Q — Part A text-based, Part B diagram-based with SVG figures) |

## Phase 3 — Achievers / HOTS + Full Papers (Weeks 11–12)

| Week | Activity |
|---|---|
| 11 | `assignments/11-achievers-hots/` ✅ — 2-and-3-step mixed problems pulling from every Phase-1 topic. Then attempt **Sample Paper 1** (untimed, open-book) to find the gaps — see `sample-papers/README.md`. |
| 12 | Attempt **Sample Papers 2–5** (Level-1), each **strictly timed to 60 minutes**, one every 1–2 days. After each paper, log every wrong answer in an error log (topic + why it was wrong) before moving to the next paper. |
| 13 (if qualified) | Work through the 3 **Level-2 topic assignments** ( `assignments/13-achievers-level2/`, `14-reasoning-level2/`, `15-geometry-level2/` ) untimed first, to rebuild each skill at the harder difficulty. Then attempt **Sample Papers 6–9** (Level-2 track), timed, once Level 1 results come in and you've qualified for Level 2 — see below. |

### Sample paper schedule (Phase 3)

| Paper | When | Mode |
|---|---|---|
| Sample Paper 1 | Week 11, end | Untimed, review-focused |
| Sample Paper 2 | Week 12, Day 1 | Timed (60 min) |
| Sample Paper 3 | Week 12, Day 3 | Timed (60 min) |
| Sample Paper 4 | Week 12, Day 5 | Timed (60 min) |
| Sample Paper 5 | Week 12, Day 7 | Timed (60 min) — treat as Level-1 dress rehearsal |
| Sample Paper 6 | After qualifying Level 1 | Timed (60 min), Level-2 difficulty |
| Sample Paper 7 | 1–2 days after Paper 6 | Timed (60 min), Level-2 difficulty |

| Paper | When | Mode |
|---|---|---|
| Sample Paper 8 | 1–2 days after Paper 7 | Timed (60 min), Level-2 difficulty |
| Sample Paper 9 | 1–2 days after Paper 8 | Timed (60 min), Level-2 difficulty |

### Level-2 topic assignments (new)

Before the Level-2 sample papers, three dedicated harder topic worksheets are available, for the areas where Level 2 gets conceptually harder (not just bigger numbers):

| Folder | Topic |
|---|---|
| `assignments/13-achievers-level2/` ✅ | Multi-step HOTS (working backwards, fraction-of-remainder, age/rate problems) |
| `assignments/14-reasoning-level2/` ✅ | Compound logical reasoning (2+ stacked clues per question) |
| `assignments/15-geometry-level2/` ✅ | Compound-shape perimeter & area (paths, frames, cut corners) |

All 9 sample papers are built out (full 35 questions, 40 marks, answer key each) — see `sample-papers/README.md`. Papers 1–5 are Level-1 difficulty; Papers 6–9 are a dedicated **Level-2 track** for students who've qualified Level 1 and are prepping for the actual, harder Level-2 exam.

## Progress tracker

Copy this checklist and tick off as you go:

- [ ] Week 1 — Number Sense *(assignment ready)*
- [ ] Week 2 — Four Operations *(assignment ready)*
- [ ] Week 3 — Factors, Multiples & Patterns *(assignment ready)*
- [ ] Week 4 — Fractions & Decimals *(assignment ready)*
- [ ] Week 5 — Measurement & Money *(assignment ready)*
- [ ] Week 6 — Time & Calendar *(assignment ready)*
- [ ] Week 7 — Geometry, Perimeter & Area *(assignment ready)*
- [ ] Week 8 — Data Handling *(assignment ready)*
- [ ] Week 9 — Reasoning: Patterns/Analogy/Coding *(assignment ready)*
- [ ] Week 10 — Reasoning: Ranking/Direction/Calendar *(assignment ready)*
- [ ] Week 11 — Achievers/HOTS + Sample Paper 1 *(assignment + paper ready)*
- [ ] Week 12 — Sample Papers 2–5, timed *(Level-1, all ready)*

- ☐ Week 13 — Level-2 topic assignments (Achievers, Reasoning, Geometry), if qualified *(all ready)*
- ☐ Week 13 — Sample Papers 6–9, timed, if qualified for Level 2 *(Level-2 track, all ready)*

(The checkboxes above track *your* progress working through the plan, not whether the content exists — every assignment and sample paper listed is already built.)

## Optional extension

Once the 12 weeks and all 6 sample papers are comfortably done (consistently ≥32/40, i.e. Level-2 cutoff range), move to `assignments/12-math-kangaroo-stretch/` ✅ — 15 Math Kangaroo-style puzzle questions (shape counting, mirror images, balance-scale logic, grid paths, cube nets, and classic brain-teasers like the "snail in the well") in 3 difficulty tiers, for an extra spatial-reasoning/creative-problem-solving stretch beyond anything the core plan covers. See the note in `01-syllabus.md` §3.

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